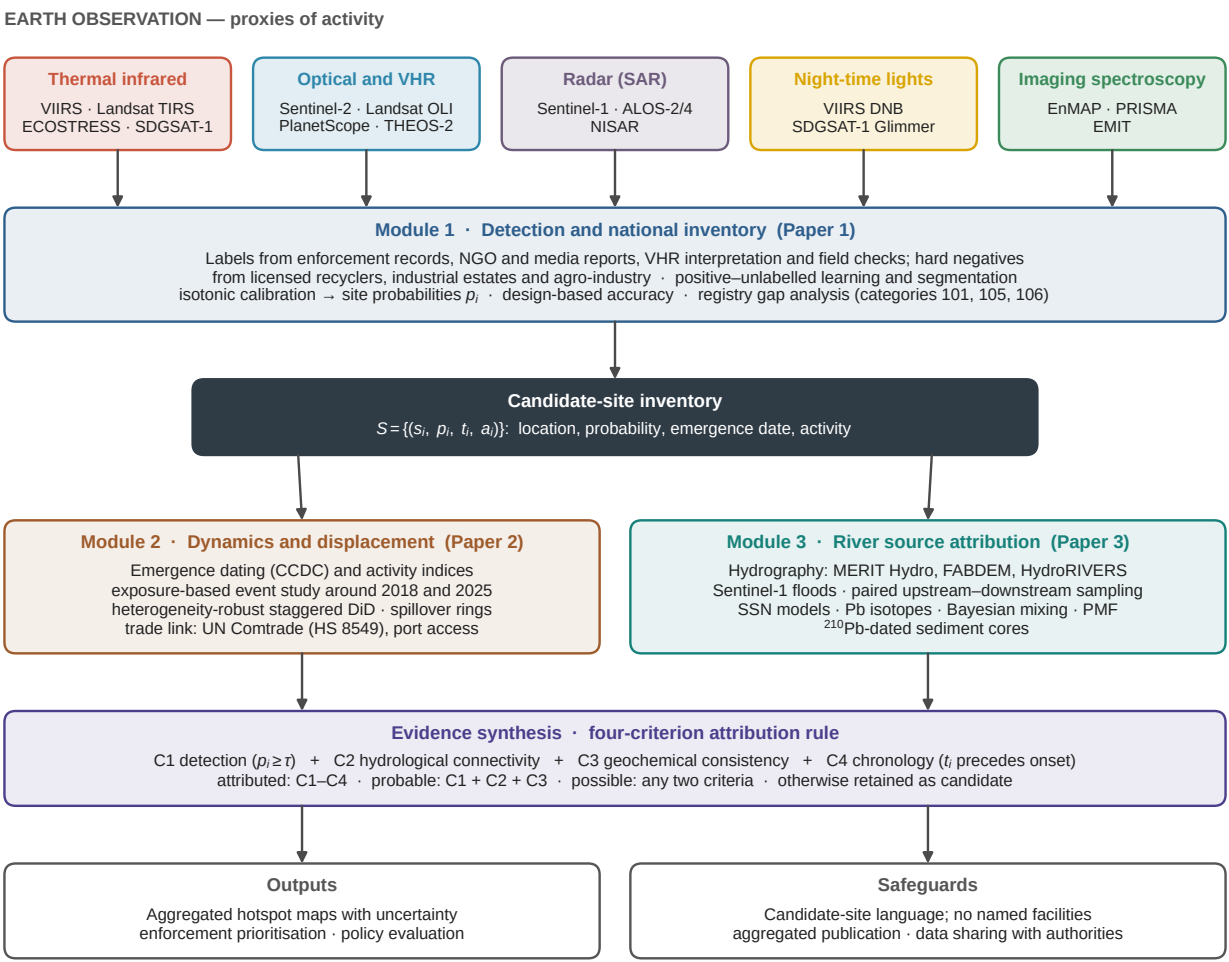
EARTH OBSERVATION — proxies of activity
Thermal infrared
VIIRS · Landsat TIRS
ECOSTRESS · SDGSAT-1
Optical and VHR
Sentinel-2 · Landsat OLI
PlanetScope · THEOS-2
Radar (SAR)
Sentinel-1 · ALOS-2/4
NISAR
Night-time lights
VIIRS DNB
SDGSAT-1 Glimmer
Imaging spectroscopy
EnMAP · PRISMA
EMIT
Module 1 · Detection and national inventory (Paper 1)
Labels from enforcement records, NGO and media reports, VHR interpretation and field checks; hard negatives
from licensed recyclers, industrial estates and agro-industry · positive–unlabelled learning and segmentation
isotonic calibration → site probabilities p_i · design-based accuracy · registry gap analysis (categories 101, 105, 106)
Candidate-site inventory
$S = \{(s_i,\ p_i,\ t_i,\ a_i)\}$: location, probability, emergence date, activity
Module 2 · Dynamics and displacement (Paper 2)
Emergence dating (CCDC) and activity indices
exposure-based event study around 2018 and 2025
heterogeneity-robust staggered DiD · spillover rings
trade link: UN Comtrade (HS 8549), port access
Module 3 · River source attribution (Paper 3)
Hydrography: MERIT Hydro, FABDEM, HydroRIVERS
Sentinel-1 floods · paired upstream–downstream sampling
SSN models · Pb isotopes · Bayesian mixing · PMF
^{210}Pb-dated sediment cores
Evidence synthesis · four-criterion attribution rule
C1 detection ($p_i \geq \tau$) + C2 hydrological connectivity + C3 geochemical consistency + C4 chronology (t_i precedes onset)
attributed: C1–C4 · probable: C1 + C2 + C3 · possible: any two criteria · otherwise retained as candidate
Outputs
Aggregated hotspot maps with uncertainty
enforcement prioritisation · policy evaluation
Safeguards
Candidate-site language; no named facilities
aggregated publication · data sharing with authorities